



Multiplexing Techniques

◆ Definition

Multiplexing is a technique used to **combine multiple signals into a single communication channel** to improve bandwidth utilization and reduce cost. The reverse process is called **demultiplexing**.

◆ 1. Frequency Division Multiplexing (FDM)

Concept:

- The available bandwidth is divided into **multiple frequency bands**
- Each signal is transmitted **simultaneously on different frequencies**

Features:

- Used for **analog signals**
- Requires **guard bands** to prevent overlap

Advantages:

- Continuous transmission
- No delay between signals

Disadvantages:

- Bandwidth wastage due to guard bands
- Susceptible to noise

Applications:

- Radio broadcasting
 - Cable TV
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◆ 2. Time Division Multiplexing (TDM)

Concept:

- Each signal is assigned a **specific time slot**
- Signals transmit **one after another in sequence**

Types:

- **Synchronous TDM** → Fixed time slots
- **Asynchronous (Statistical) TDM** → Dynamic slots based on demand

Advantages:

- Efficient bandwidth utilization
- No need for guard bands

Disadvantages:

- Requires synchronization
- Delay may occur

Applications:

- Telephone systems
 - Digital communication
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◆ 3. Wavelength Division Multiplexing (WDM)

Concept:

- Used in **optical fiber communication**
- Multiple signals are transmitted using **different wavelengths (colors)** of light

Types:

- **CWDM (Coarse WDM)**
- **DWDM (Dense WDM)**

Advantages:

- Very high data transmission rates
- Efficient use of optical fiber

Disadvantages:

- High cost
- Complex technology

Applications:

- Fiber optic communication
 - Internet backbone
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4. Code Division Multiplexing (CDM / CDMA)

Concept:

- All users share the **same frequency and time**
- Each signal is encoded with a **unique code**

Features:

- Based on **spread spectrum technique**

Advantages:

- High security
- Resistant to interference and noise

Disadvantages:

- Complex system
- Requires strict synchronization

Applications:

- Mobile communication (3G)
 - GPS systems
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◆ Comparison Table

Technique	Basis	Key Idea	Signal Type
FDM	Frequency	Different frequencies	Analog
TDM	Time	Different time slots	Digital
WDM	Wavelength	Different light wavelengths	Optical
CDM	Code	Unique codes	Digital

🧠 Quick Summary (Exam Trick)

👉 **F → T → W → C = Frequency, Time, Wavelength, Code**